QUESTION NO -1:

Aim

To develop a Python program to represent the distribution of grades obtained by 40 students using a pie chart and a bar chart, identify the maximum and minimum grades by student count, and calculate the combined percentage of students who secured grades A and S.

Algorithm

1. Start the program.
2. Store the grades F, E, D, C, B, A and S in a list.
3. Store the corresponding number of students as 3, 4, 5, 8, 9, 7 and 4.
4. Calculate the total number of students and the percentage for each grade.
5. Draw a pie chart showing the percentage distribution of grades.
6. Display grade labels and percentage values on the pie chart.
7. Draw a bar chart comparing the number of students in each grade.
8. Label the X-axis as Grades and the Y-axis as Number of Students.
9. Display the exact number of students above each bar.
10. Identify the grades with the maximum and minimum number of students.
11. Calculate the combined percentage of students who secured grades A and S.
12. Display the results.
13. Stop the program.

CODE:

```
import matplotlib.pyplot as plt


# Store grades and number of students
grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]
```

```
# Calculate total students
total = sum(students)

print("Total number of students:", total)
print("\nGrade-wise student details:")

# Calculate and display percentages
for i in range(len(grades)):
    percentage = (students[i] / total) * 100
    print(grades[i], ":", students[i],
          "students -", round(percentage, 2), "%")

# Pie chart
plt.figure(figsize=(7, 6))
plt.pie(
    students,
    labels=grades,
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Percentage Distribution of Student Grades")
plt.legend(grades, title="Grades")
plt.show()

# Bar chart
plt.figure(figsize=(8, 5))
bars = plt.bar(grades, students)
```

```
plt.xlabel("Grades")
plt.ylabel("Number of Students")
plt.title("Number of Students in Each Grade")

# Display values above bars
plt.bar_label(bars)

plt.show()

# Find maximum and minimum
max_count = max(students)
min_count = min(students)

max_grades = [
    grades[i] for i in range(len(grades))
    if students[i] == max_count
]

min_grades = [
    grades[i] for i in range(len(grades))
    if students[i] == min_count
]

print("\nMaximum students:", max_grades, "-", max_count)
print("Minimum students:", min_grades, "-", min_count)

# Combined percentage of A and S
```

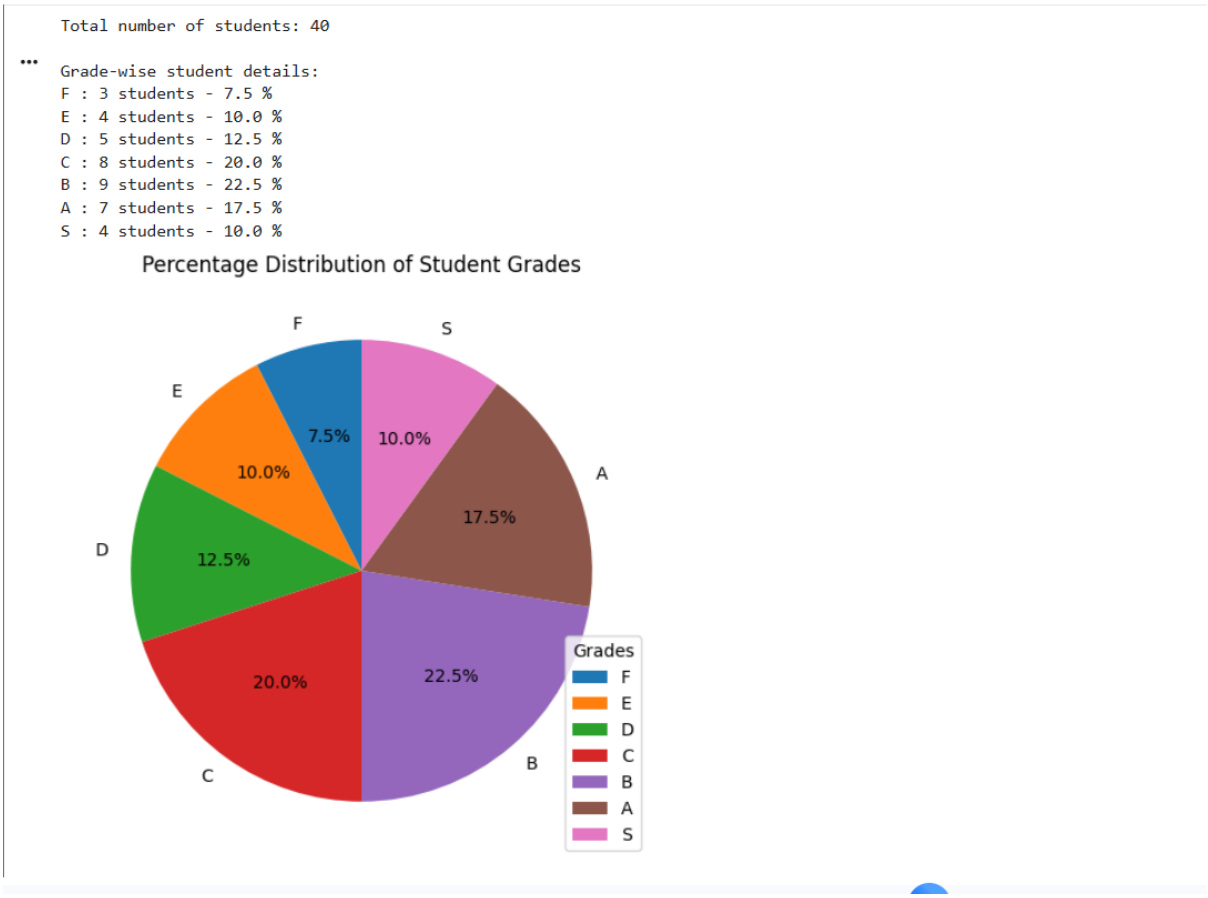
```
combined = students[5] + students[6]
percentage = (combined / total) * 100

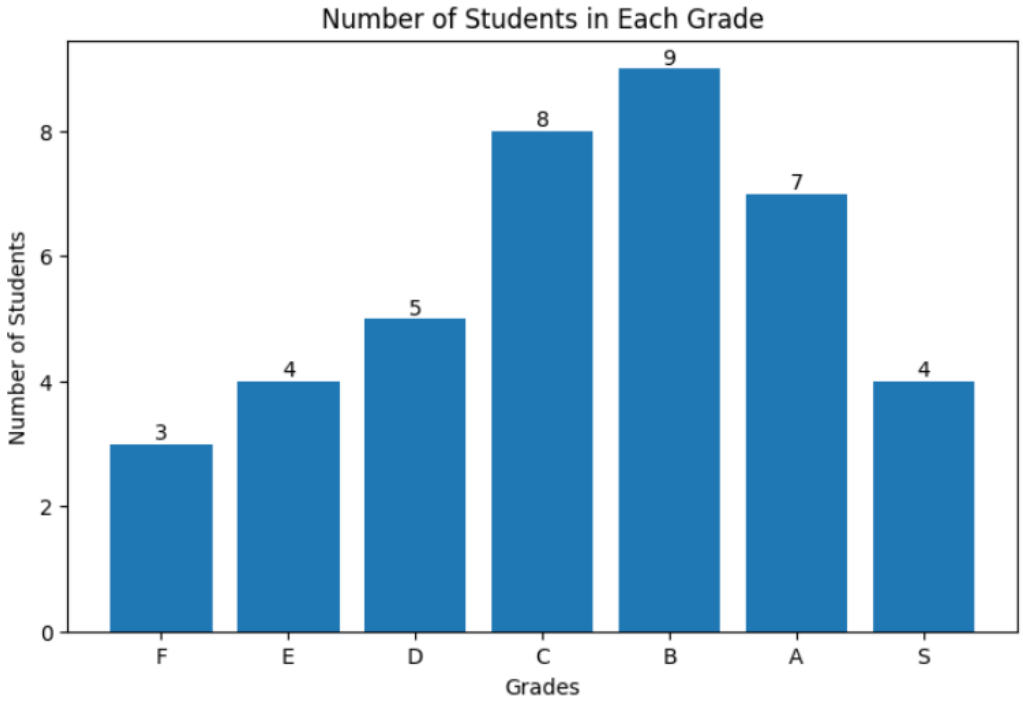
print("Students with grades A and S:", combined)
print("Combined percentage of A and S:",
      round(percentage, 2), "%")
```

Expected output :

```
Total number of students: 40

Grade-wise student details:
F : 3 students - 7.5 %
E : 4 students - 10.0 %
D : 5 students - 12.5 %
C : 8 students - 20.0 %
B : 9 students - 22.5 %
A : 7 students - 17.5 %
S : 4 students - 10.0 %
```






```
Maximum students: ['B'] - 9
Minimum students: ['F'] - 3
Students with grades A and S: 11
Combined percentage of A and S: 27.5 %
```